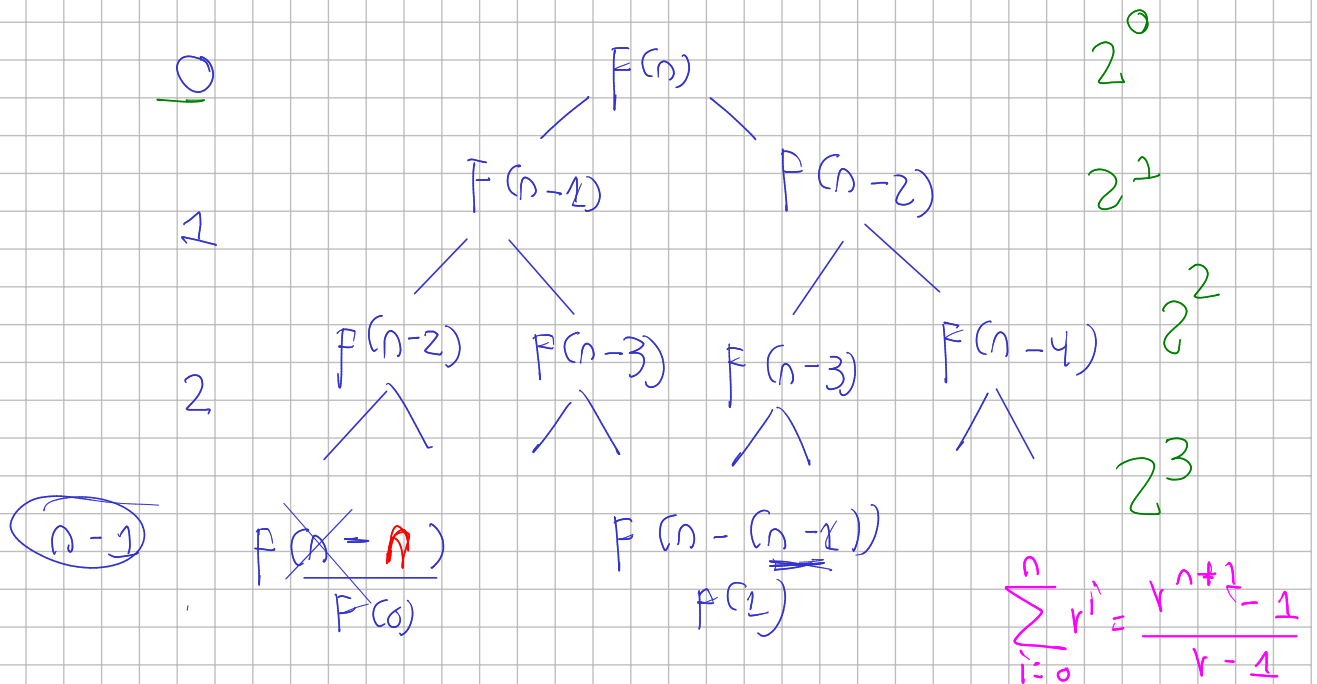


$$F(n) = \begin{cases} 0 & n=0 \\ 1 & n=1 \\ F(n-1) + F(n-2) \end{cases}$$



$$2^0 + 2^1 + 2^2 + 2^3 + \dots + 2^{n-1}$$

$$\sum_{k=0}^{n-1} 2^k = \frac{2^n - 1}{2 - 1} = 2^n - 1 = O(2^n)$$